

Refugee Inflows and Native Parity Progression

Evidence from Turkey

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1. **Turkey hosts the world's largest refugee population.** $\approx 3.6\text{M}$ Syrians in Turkey by 2018, out of $\approx 5.3\text{M}$ registered across host countries.
2. **Refugees have much higher fertility than natives.** Syrian refugee women's TFR sharply rebounded after resettlement and remains well above Turkish native TFR; Syrians also marry earlier and reach higher mean parity at every age window.
3. **Native fertility is low and falling.** Turkish TFR fell from 2.38 in 2001 to 1.51 in 2023, well below the 2.10 replacement threshold and below the OECD average.

A *large* inflow with a *large* host–migrant fertility differential — an **ideal setting** to examine the potential impact of refugees on native fertility.

Research question. How does the influx of Syrian refugees affect the fertility decisions of native Turkish women? We study both **marriage** and **parity-by-age** outcomes (1–5 children).

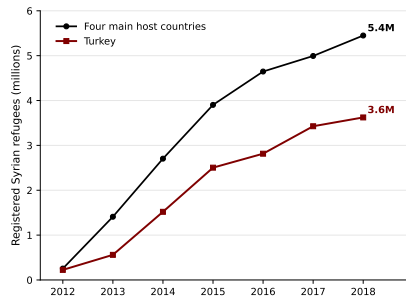
Fertility gap

Table 1

Table 2

Background — The Syrian Civil War and Displacement into Turkey

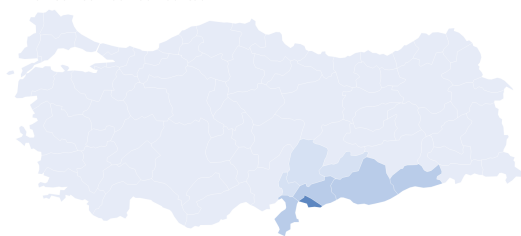
- **March 2011:** an uprising in Syria escalates into civil war; over half of Syria's prewar population is eventually displaced.
- **Turkey adopts an “open door” policy** (Oct 2011). Registered Syrians grow from $\approx 224\text{k}$ (2012) to $\approx 3.6\text{M}$ (2018): $\approx 4.4\%$ of Turkey's population, the world's largest refugee stock.
- **Legal status: “temporary protection,” not refugee status.** No formal asylum track (Turkey retains the 1951 Convention's geographic limitation); registration grants free public health care and schooling.
- **No formal labor market access until 2016.** Work permits arrive in Jan 2016 with take up of $\approx 1\text{--}2\%$, so employment is overwhelmingly **informal**.
- **From camps to cities.** By 2018, over 90% live outside camps.



Registered Syrian refugees: total across the four main host countries (Turkey, Lebanon, Jordan, Iraq) and Turkey alone. Sources: host country registries; DGMM.

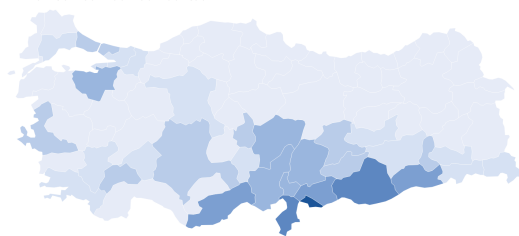
Background — Where Refugees Settled

Percentage of Syrian Refugees to Natives (2013)



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Percentage of Syrian Refugees to Natives (2018)

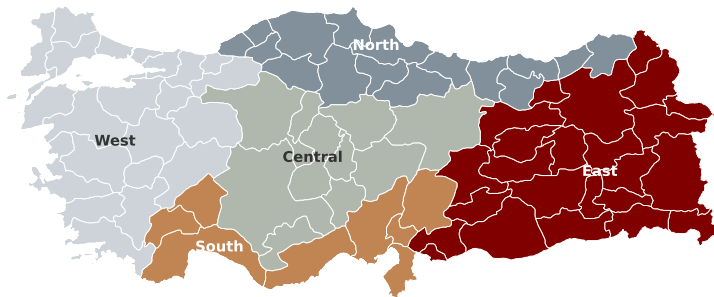


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Registered Syrians / native population (our exposure measure). 2018: **Kilis 89%**, Hatay 28%, Şanlıurfa 23%, Gaziantep 20%; Istanbul hosts the largest *absolute* number ($\approx 560k$, 3.7%); the **median province is at 0.6%**, and 6 in 10 provinces are below 1%.

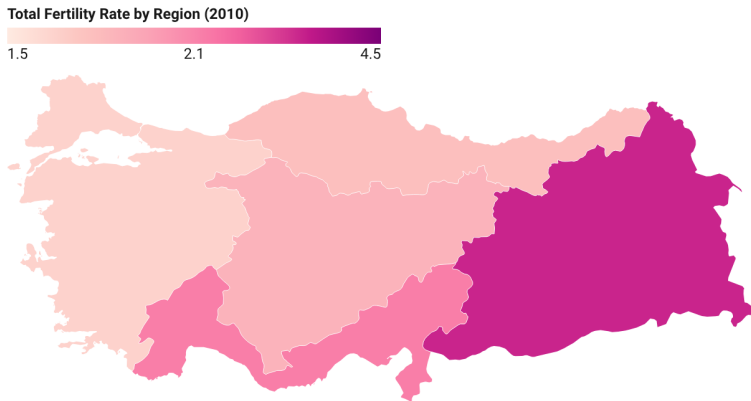
Arrival concentrated along the border, then diffusion westward. Differences across provinces remain **huge and persistent**; this geographic variation, activated over time, is the identifying variation.

Background — Turkey's Five Regions



The five regions in our study: **West**, **Central**, **South**, **North**, and **East**. They differ sharply in geography, culture, and demography. Refugee exposure is concentrated in the two regions shown in warm tones: the **East** (poorer, more rural, home to most Kurdish and Arabic speakers, highest fertility) and the **South**. The **West** is richer, urban, and has the lowest fertility. These persistent differences matter for **interpretation**, since who is exposed shapes which mechanisms are plausible, and for **identification**, since each region gets its own time trend.

Background — Fertility Differs Sharply Across These Regions

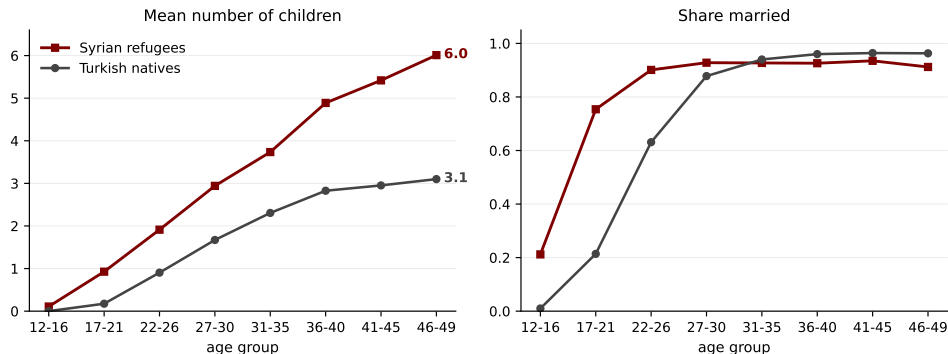


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Total fertility rate by region in 2010, the year before the inflow: from about 1.5 in the **West** to well above replacement in the **East**. The refugees settled into the **highest fertility part of the country**.

- **An informal labor market.** Roughly a third of Turkish employment is unregistered; refugees, lacking permits, work almost entirely informally. Established result: the inflow displaced natives from *informal* jobs, with **women hit hardest** (Del Carpio & Wagner 2015; Ceritoglu et al. 2017; Aksu, Erzan & Kırdar 2022).
- **A fertility gap.** Native TFR ≈ 2.3 and falling over the study window; Syrian refugees in Turkey have TFR ≈ 5.3 (TDHS 2018): refugees are younger, marry earlier, and reach every parity sooner.
- **Ethnolinguistic geography.** The border provinces with the highest exposure are home to Turkey's **Kurdish** ($\approx 16\%$ of women) and **Arabic** ($\approx 2\%$) mother tongue minorities; most refugees are Arabs, many are Kurds, from adjacent northern Syria. **Cultural and linguistic proximity is highest exactly where exposure is highest.**

Background — The Fertility Gap, by Age



Source: TDHS 2013 and 2018; Syrian refugees from the 2018 round (first nationally representative TDHS sample of Syrians). At every age Syrian women have roughly **twice as many children**; at ages 17–21, **three quarters are married** versus one in five natives.

Effects of the same inflow on *native labor markets* (*Labour Economics*; HLFS microdata 2004–2015).
The key input for interpreting our results:

Substitute workers (informal) lose.

- Native **informal employment falls** sharply: temporary wage workers, workers with less education, agriculture and construction.
- Total **female employment falls**, driven by the loss of part time and informal jobs.

Complementary workers (formal) gain.

- Native men shift **from informal into formal** work; men's formal wage employment and **formal sector wages rise** (especially manufacturing).
- **No decline in total male employment** and no decline in average wages of men or women.

Through a fertility lens. The shock lowered the **opportunity cost of women's time** while leaving **household income intact**: the classic substitution effect configuration that predicts *more* fertility, concentrated where informal work dominates.

Main Findings.

- **No effect on marriage or early-parity fertility** (parities 1–3, all ages).
- 2SLS: large positive estimates at **higher parities (4–5)** in the **late 20s / 30s**.
- Evidence of an increase in higher-parity attainment concentrated in the **Arabic/Kurdish mother-tongue** subgroup.
- Weaker, suggestive evidence of the same pattern in **lower-educated** women and the **high-exposure East/Southeast region**.

Mechanism and consistency check.

- **Labor-market / opportunity-cost** (Aksu, Erzan, Kırdar 2022): refugee inflow reduced *native female* employment but not male employment $\Rightarrow$ lower opportunity cost of an extra child without an income hit.
- **Stated ideal- n** shifts upward at the same upper thresholds and in the same subgroups where realised higher-parity fertility responds — a consistency check on the labor-market reading, not a separate channel.

Immigration → **native fertility: prior evidence (high-income settings).**

- **Childcare-cost channel.** Immigration lowers the price of household / care services and raises fertility of *highly-educated* natives. Furtado (2016); Furtado et al. (2015); Romiti (2018); Farré, González & Ortega (2011); Forlani et al. (2021).
- **Labor-market / opportunity-cost channel.** Female-targeted labor shocks move fertility more than male-targeted ones: Huttunen & Kellokumpu (2016); Schaller (2016); Del Bono, Weber & Winter-Ebmer (2015).

Contribution.

1. **Demographic spillovers of refugee inflows.** Prior work on the Syrian shock studies labor, housing, prices, crime, health, education; we document that large refugee inflows can alter *native fertility* trajectories in economically meaningful ways.
2. **Migration → fertility in a middle-income context.** In contrast to the childcare-cost evidence from rich countries, we find *no* positive response among highly-educated natives; the response is concentrated in lower-education and culturally-proximate groups — the dominant mechanism in developed economies does not generalise here.
3. **Parity-specific lens on the fertility transition.** Tracking the full ladder $\mathbb{1}\{\text{parity} \geq p \text{ by age } a\}$ localizes the response at *higher parities among vulnerable groups* — a channel through which migration can widen within-host demographic inequality.

TDHS (Turkey Demographic and Health Survey; Hacettepe University, part of the international DHS program): nationally representative microdata on women with rich **fertility histories** and **migration histories** for responding women.

Main data: TDHS **2013 & 2018**; woman $\times$ age **pseudo-panel** from age 12 to survey age; province assigned by **2011 residence** (pre-shock).

Cumulative parity outcomes (absorbing-state indicators):

$$Y_{ija}^{(p)} = \mathbb{1}\{\text{parity}_i \geq p \text{ by age } a\}, \quad p \in \{1, 2, 3, 4, 5\}, \quad a \in \{16, 20, 24, 28, 32, 36\}.$$

Plus the marriage analogue $\mathbb{1}\{\text{ever married by age } a\}$.

Exposure: average refugee-to-native ratio over the woman's age window 12 to a :

$$\bar{R}_{ja} = \frac{1}{a - 11} \sum_{s=12}^a R_{j, t(s,i)}.$$

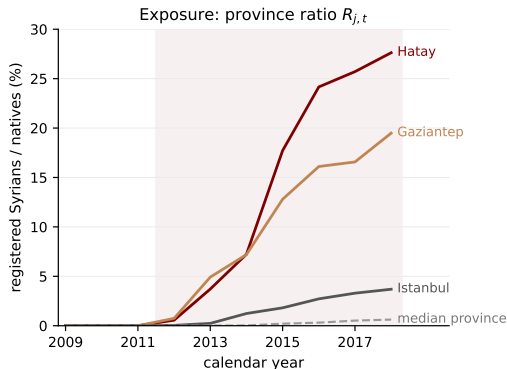
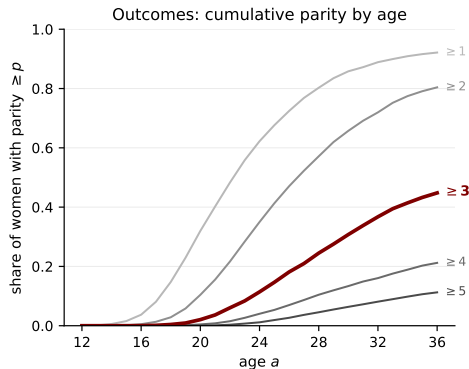
Interpretation. A 1 pp. higher *average annual* ratio over ages 12– a raises $\Pr(\text{parity} \geq p \text{ by } a)$ by $\hat{\beta}^{(p,a)}$ pp.

[Sample details](#)

[Data sources](#)

The Outcome and the Exposure, in the Data

Left: the outcome family: each **(parity, age)** cell of the grid is a point on these curves. **Right:** the exposure: province ratio paths; $\bar{R}_{ja}$ **averages the woman's province path** over ages 12 to a .



TDHS 2013+2018 estimation sample, weighted; Ministry of Interior / AFAD province ratios.

Worked example

Reconstructing the Pseudo-Panel

Cross-sectional record (TDHS interview):

ID	Birth Yr	Childhood Prov	Move Yr	Dest. Prov	Birth 1	Birth 2
10001	1990	Adana	2009	Mardin	2010	2013

⇓ expanded into woman-by-year panel ⇓

Pseudo-panel:

ID	Age	Year	Province	Gave Birth 1	Gave Birth 2
10001	12	2002	Adana	0	0
⋮	⋮	⋮	⋮	⋮	⋮
10001	20	2010	Mardin	1	0
10001	23	2013	Istanbul	1	1

Outcomes are collapsed to one row per (i, a) for each age cutoff; exposure $\bar{R}_{ja}$ averaged over age 12– a .

Identification — Quasi-Exogenous Distance Variation

Challenge. Refugee location is endogenous: internal migration responds to local labor-market conditions; by 2018 refugees had spread well beyond border provinces.

Strategy. Follow Aksu, Erzan & Kirdar (2022): instrument $\bar{R}_{ja}$ with a shift-share variable based on geographic distance from Syrian to Turkish provinces.

$$Z_{n,t} = \sum_{s=1}^{13} \frac{(1/d_{s,n}) \pi_s}{\sum_{X \in \{T,L,J,I\}} (1/d_{s,X})} \times T_t = B_n \times T_t.$$

π_s : pre-war population share of Syrian province s ; $d_{n,s}$: distance from Syrian s to Turkish n ; T_t : aggregate Syrian refugee stock across the four host countries (Turkey, Lebanon, Jordan, Iraq).

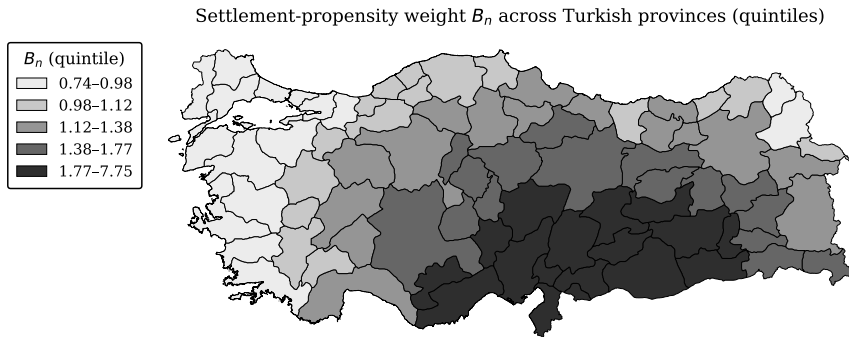
Where the variation comes from.

- B_n is a time-invariant, geography-based intensity of refugee exposure for province n : provinces closer to Syrian provinces have higher B_n .
- **Exclusion restriction.** Conditional on FE and trend controls, B_n is uncorrelated with pre-shock fertility trends among Turkish natives.

T_t over time

B_n map

Settlement-Propensity Weight B_n across Provinces



For each woman i in province j , we model whether parity p has been reached by age a :

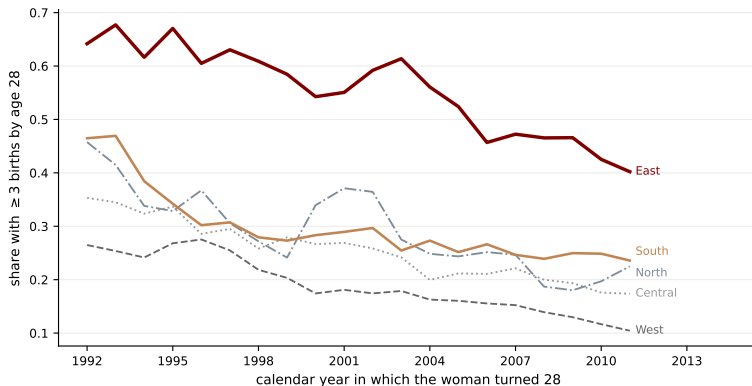
$$Y_{ija}^{(p)} = \alpha + \beta^{(p,a)} \bar{R}_{ja} + X_i' \Gamma + \delta_j + \mu_{t(a,i)} + \theta_{g(j)} \cdot t(a, i) + \lambda_{l(i)} \cdot t(a, i) + \varepsilon_{ija},$$

with $\bar{R}_{ja}$ instrumented by $\bar{Z}_{ja} = B_j \times \bar{T}_a$.

- $\beta^{(p,a)}$ — coefficient of interest: effect of a 1 pp. higher *average annual* refugee-to-native ratio on $\Pr(\text{parity} \geq p \text{ by age } a)$.
- X_i' — individual controls: survey wave, childhood urban/rural, mother's education, mother tongue.
- δ_j — province FE; $\mu_{t(a,i)}$ — year-at-age- a FE ($\equiv$ birth-cohort FE at fixed a).
- $\theta_{g(j)} \cdot t$ — **five-region**-specific linear trends.
- $\lambda_{l(i)} \cdot t$ — **mother-tongue**-specific linear trends (Turkish / Kurdish / Arabic).
- Trends start in 1987; SEs clustered at province; DHS sampling weights.

Geography, for reference: 81 provinces; the five regions shown earlier; NUTS-1 / NUTS-2 = 12 / 26 Eurostat statistical regions (used in robustness and in the hazard design).

Why Do Regions Need Their Own Time Trends?



Our outcome before the shock: share of women with ≥ 3 births by age 28, by region and by the calendar year the woman turned 28 (TDHS 2013+2018 birth histories, weighted, smoothed, cohorts reaching 28 by 2011). The **East** sits far above the rest and declines on its own path; regions **converge at different rates**. The inflow landed in exactly the regions with the highest and most distinctly trending fertility: without region trends, this differential evolution loads onto B_n .

[More outcomes](#)

Placebo — Cumulative Design: $Y \sim B_n \times t$ on Pre-Shock Cohorts

On the pre-shock subsample (cohort-age window ends before 2012), replace the shift-share instrument $\bar{Z}_{ja}$ with $B_j \times t(a, i)$ — a linear placebo interaction in calendar time. Under the null of no pre-trends in B_j , $\beta(p, a)$ should be zero.

$$Y_{ija}^{(p)} = \alpha + \gamma^{(p,a)}(B_j \times t(a, i)) + X_i' \Gamma + \delta_j + \mu_{t(a,i)} + \theta_{g(j)} t + \varepsilon_{ija}.$$

Outcome	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
Ever Married	0.0003 (0.0005) [0.84]	0.0010 (0.0007) [0.47]	0.0012* (0.0007) [0.47]	0.0008** (0.0004) [0.38]	0.0011* (0.0006) [0.47]	-0.0001 (0.0007) [0.90]
Ever Gave Birth to One Child	0.0005* (0.0003) [0.77]	0.0011 (0.0008) [0.92]	0.0007 (0.0010) [1.00]	0.0011** (0.0005) [0.62]	0.0011** (0.0005) [0.74]	0.0001 (0.0007) [1.00]
Ever Gave Birth to Two Children	0.0001 (0.0002) [1.00]	-0.0001 (0.0004) [1.00]	0.0009 (0.0006) [0.91]	0.0018*** (0.0007) [0.48]	0.0015 (0.0010) [0.91]	0.0006 (0.0011) [1.00]
Ever Gave Birth to Three Children	0.0000 (0.0000) [0.92]	-0.0000 (0.0002) [1.00]	0.0004 (0.0007) [1.00]	0.0000 (0.0008) [1.00]	-0.0007 (0.0008) [0.99]	-0.0024* (0.0013) [0.79]
Ever Gave Birth to Four Children	— (—)	— (—)	0.0002 (0.0004) [1.00]	0.0011 (0.0009) [0.92]	0.0005 (0.0023) [1.00]	-0.0005 (0.0022) [1.00]
Ever Gave Birth to Five Children	— (—)	— (—)	0.0000 (0.0002) [1.00]	0.0002 (0.0005) [1.00]	0.0012 (0.0014) [0.99]	0.0018 (0.0022) [0.99]
Observations	12,355	11,560	10,226	8,194	6,063	4,098

Near-zero $\hat{\gamma}^{(p,a)}$. Stars = raw inference; **[brackets]** = Romano–Wolf stepdown p -values across the 32-cell family. No cell rejects after Romano–Wolf.

What is RW?

First stage

NUTS-1 placebo

NUTS-2 placebo

Main Results — Cumulative Design, Romano–Wolf p -values

Outcome	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
Ever Married	-0.047 (0.217) [1.00]	-0.572 (0.600) [0.90]	0.513 (1.689) [1.00]	-0.013 (1.047) [1.00]	0.220 (0.920) [1.00]	-0.299 (1.241) [1.00]
Ever Birth ≥ 1	0.010 (0.127) [1.00]	0.123 (0.523) [1.00]	-0.043 (1.968) [1.00]	0.923 (1.342) [1.00]	-1.515 (1.789) [0.99]	0.184 (1.371) [1.00]
Ever Birth ≥ 2	0.028 (0.041) [1.00]	0.305 (0.546) [1.00]	-0.151 (0.746) [1.00]	0.613 (1.906) [1.00]	1.740 (3.419) [1.00]	-0.845 (1.189) [1.00]
Ever Birth ≥ 3	0.008 (0.005) [0.88]	0.382 (0.248) [0.88]	-0.371 (0.565) [1.00]	1.250 (1.559) [1.00]	2.893 (2.700) [0.98]	1.062 (1.810) [1.00]
Ever Birth ≥ 4	– (–)	– (–)	0.250 (0.340) [1.00]	1.905** (0.809) [0.52]	2.246 (1.415) [0.86]	4.252** (1.963) [0.61]
Ever Birth ≥ 5	– (–)	– (–)	0.313 (0.232) [0.92]	1.392* (0.803) [0.81]	1.414 (0.912) [0.88]	4.133** (1.856) [0.58]
Observations	14,354	13,484	12,241	10,240	8,245	6,110

Stars = raw inference; **[brackets]** = Romano–Wolf stepdown p -values across the 32-cell family. Positive raw effects at parities 4–5, ages 28+; no cell rejects after Romano–Wolf in the pooled sample.

What is RW?

4-wave EM

OLS

Robustness

Lower-Educated Women

Outcome	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
<i>Lower education (no schooling / primary)</i>						
Ever Married	-0.185 (0.303) [0.95]	-0.872 (1.285) [0.95]	2.048 (1.824) [0.86]	-0.059 (1.529) [1.00]	0.189 (1.035) [1.00]	-0.203 (1.190) [1.00]
Ever Birth ≥ 1	-0.023 (0.187) [1.00]	0.048 (1.249) [1.00]	0.291 (2.443) [1.00]	1.303 (1.957) [1.00]	0.045 (1.274) [1.00]	0.558 (1.331) [1.00]
Ever Birth ≥ 2	0.003 (0.057) [1.00]	0.261 (1.636) [1.00]	-1.116 (1.682) [1.00]	1.691 (1.645) [0.98]	1.602 (2.743) [1.00]	1.986 (1.635) [0.96]
Ever Birth ≥ 3	0.004 (0.007) [1.00]	0.738 (0.765) [0.99]	-0.637 (1.210) [1.00]	2.823 (2.238) [0.96]	3.261 (2.446) [0.94]	4.689*** (1.697) [0.33]
Ever Birth ≥ 4	— (—)	— (—)	-0.067 (0.679) [1.00]	3.175** (1.294) [0.45]	2.968*** (1.053) [0.32]	6.658*** (2.370) [0.32]
Ever Birth ≥ 5	— (—)	— (—)	0.351 (0.414) [0.99]	1.787 (1.284) [0.93]	1.893 (1.160) [0.85]	5.619*** (1.860) [0.26]
Observations	9,361	9,132	8,729	7,633	6,348	4,897

Those with primary school or less. Raw point estimates large and positive at parities **4–5, ages 28–36**; no cell rejects after Romano–Wolf (smallest RW $p \approx 0.26$) — we read this as suggestive evidence.

Baseline rates

EM analog

Higher-educated

Outcome	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
<i>Arabic / Kurdish mother tongue</i>						
Ever Married	0.556* (0.323) [0.53]	0.520 (1.759) [0.96]	0.585 (2.478) [0.96]	-4.025* (2.239) [0.52]	1.933 (2.238) [0.87]	0.859 (1.146) [0.87]
Ever Birth ≥ 1	0.336* (0.177) [0.73]	1.821 (1.593) [0.95]	-1.754 (2.323) [0.99]	-1.585 (2.396) [0.99]	2.441 (2.770) [0.98]	0.762 (1.811) [0.99]
Ever Birth ≥ 2	0.047 (0.073) [0.99]	1.806 (1.166) [0.82]	1.433 (1.838) [0.99]	1.460 (2.987) [0.99]	4.418 (3.288) [0.89]	1.994 (1.401) [0.86]
Ever Birth ≥ 3	-0.019 (0.026) [0.99]	0.647 (1.229) [0.99]	-0.869 (1.230) [0.99]	12.305*** (2.469) [0.04]	6.019* (3.479) [0.78]	2.569* (1.522) [0.79]
Ever Birth ≥ 4	- (-)	- (-)	1.201 (1.318) [0.98]	8.119*** (1.752) [0.05]	2.024 (3.811) [0.99]	8.244*** (1.788) [0.05]
Ever Birth ≥ 5	- (-)	- (-)	0.394 (1.017) [0.99]	3.155 (2.450) [0.92]	4.719 (3.347) [0.86]	5.038** (2.366) [0.64]
Observations	3,145	2,785	2,415	1,951	1,489	1,055

Evidence of a higher-parity increase among Arabic / Kurdish mother-tongue women: cells reject under Romano–Wolf at conventional thresholds — parity $\geq 3@28$ ($p^{\text{RW}} = 0.04$), parity $\geq 4@28$ ($p^{\text{RW}} = 0.05$), parity $\geq 4@36$ ($p^{\text{RW}} = 0.05$). Turkish-mother-tongue women: null.

High-Exposure Border Region (E/SE)

Outcome	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
<i>High-exposure border region (East / Southeast)</i>						
Ever Married	-0.245 (0.358) [0.87]	-1.691*** (0.646) [0.27]	-2.591 (1.976) [0.67]	-0.913 (1.656) [0.87]	-0.430 (0.761) [0.87]	1.565* (0.855) [0.50]
Ever Birth ≥ 1	-0.023 (0.154) [1.00]	-0.404 (0.478) [0.98]	-3.030* (1.569) [0.65]	1.446 (2.261) [0.99]	0.339 (1.428) [1.00]	1.989* (1.185) [0.70]
Ever Birth ≥ 2	0.021 (0.052) [1.00]	1.425*** (0.471) [0.32]	1.283 (0.934) [0.83]	0.470 (2.126) [1.00]	5.495** (2.469) [0.56]	0.330 (1.105) [1.00]
Ever Birth ≥ 3	-0.015 (0.019) [0.98]	0.339 (0.429) [0.98]	-0.050 (0.599) [1.00]	5.112** (2.045) [0.49]	9.121*** (2.816) [0.29]	4.400** (1.763) [0.49]
Ever Birth ≥ 4	– (–)	– (–)	0.229 (0.575) [1.00]	2.987** (1.283) [0.55]	6.751*** (1.698) [0.15]	5.753** (2.491) [0.55]
Ever Birth ≥ 5	– (–)	– (–)	-0.014 (0.427) [1.00]	1.875** (0.934) [0.62]	4.148*** (1.390) [0.33]	6.594* (3.499) [0.65]
Observations	4,002	3,522	3,022	2,404	1,851	1,311

East / Southeast (highest B_n , ~55% Kurdish-speaking). Largest raw point estimates at parities **3–5, ages 24+**; no cell rejects after Romano–Wolf, so we read these as suggestive evidence. Rest-of-Turkey is null.

Baseline rates EM analog Rest of Turkey

A Second Design — The Parity Progression Hazard

The same panel, read as a flow. Alongside the cumulative stock we estimate the **decision margin**: does exposure *this year* change the chance of the *next* birth?

For each woman-year in which i is **at risk** for the p -th birth (has $p - 1$ children, not yet p):

$$\Pr(p\text{-th birth at age } a) = \beta_p R_{j,t} + \delta_j + \mu_t + \phi_a + X_i' \Gamma + \theta_{n(j)} \cdot t + \lambda_{l(i)} \cdot t + \varepsilon,$$

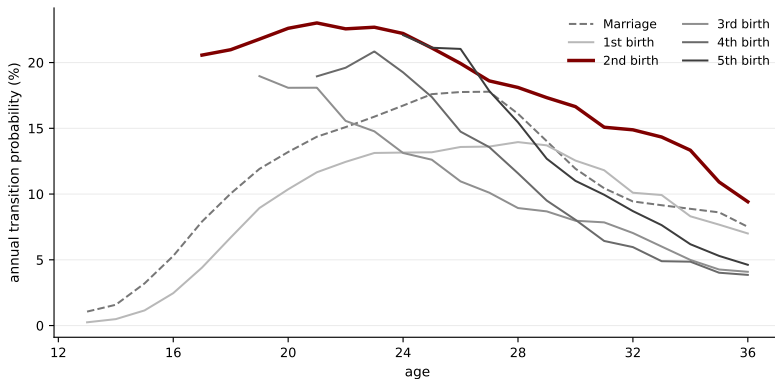
with the **contemporaneous** ratio $R_{j,t}$ instrumented by $Z_{j,t} = B_j \times T_t$.

- Same TDHS pseudo panel, kept as woman by year observations (ages 12–36); marriage + five birth transitions, each with its own risk set.
- **Six transitions instead of 32 cells**, so the multiple testing burden collapses; Romano–Wolf is applied across the 5 birth transitions (marriage is its own family).
- **Trends**: NUTS-2 (θ) and mother tongue (λ) linear trends, the set chosen *by* the hazard placebo (appendix). Province, calendar year, and age FE; SE clustered at province; DHS weights.
- The two designs answer one question from two sides: the cumulative estimate, if real, should be the **time integral** of this hazard.

Hazard placebo

The Hazard Rate, in the Data

Each curve: the share of **at risk** women making the transition at each age (marriage: never married women; p -th birth: women with exactly $p - 1$ children).



TDHS 2013+2018 person years, weighted, smoothed; ages with at least 250 observations per risk set. The **second birth** is the fastest transition, above 20% per year through the early 20s. The regression asks whether the **province ratio in a given year** shifts these annual probabilities.

Hazard Results — No Transition Responds

Full native sample; 2SLS of each transition on the contemporaneous ratio:

	Marriage	1st birth	2nd birth	3rd birth	4th birth	5th birth
2SLS $\hat{\beta}$	−0.02 (0.09)	−0.08 (0.08)	0.27 (0.27)	0.33*** (0.11)	0.08 (0.11)	−0.01 (0.15)
[RW p]	[0.82]	[0.77]	[0.77]	[0.09]	[0.77]	[0.95]
baseline $\bar{h}$	0.084	0.067	0.196	0.094	0.095	0.119
first stage F	52	54	102	123	137	102

The flow design confirms the pooled null: no transition responds after Romano–Wolf (the 3rd birth is significant on raw inference, RW $p = .09$); first stages are strong throughout. One subgroup does move: next slide.

Hazard placebo Ever married sample

	Marriage	1st birth	2nd birth	3rd birth	4th birth	5th birth
2SLS $\hat{\beta}$	0.08 (0.12)	-0.11 (0.11)	1.39*** (0.33)	0.63* (0.33)	0.13 (0.20)	-0.09 (0.26)
[RW p]	[0.58]	[0.68]	[0.03]	[0.23]	[0.77]	[0.77]
baseline $\bar{h}$	0.090	0.073	0.343	0.221	0.196	0.189
first stage F	32	29	37	39	148	37

The **second birth hazard** rises by ≈ 1.4 pp per 1 pp ratio (on a 34% baseline) and **survives Romano-Wolf** ($p = .03$); by age band, the response is concentrated at **ages 15–24** and vanishes after 25.

What this means. Among Kurdish/Arabic women who already have one child, exposure makes the **second child arrive earlier in life**: the effect shows up at ages 15 to 24 and is gone after 25. It does **not** make families larger: transitions to third, fourth, and fifth births are unchanged, and total births in a year do not rise. In short, **the same second child arrives at, say, 22 instead of 25, and completed family size stays where it was.**

Demographers call this a tempo (timing) effect rather than a quantum (family size) effect.

Age bands

Mechanism I — Labor-Market / Opportunity-Cost Channel

For $Y_i \in \{W_i, H_i\}$ (woman currently employed; husband worked last 7 days), on the TDHS women whose fertility we study, with $\bar{R}$ instrumented by $B_{j(i)} \bar{T}_{a_i^*}$:

$$Y_i = \beta \bar{R}_{j(i), a_i^*} + \alpha_{j(i)} + \mu_{t_i} + X_i' \gamma + \sum_r \lambda_r D_{it}^r + \varepsilon_i.$$

Subgroup	(1) Woman employed	(2) Husband worked 7d
Pooled	-2.7**	+2.0*
Lower education	-4.8***	+2.8**
Arabic / Kurdish mother tongue	-3.9**	≈ 0
High-exposure border region (E/SE)	-2.2*	≈ 0

Gender-asymmetric pattern: female employment falls in the pooled sample and in every subgroup where the higher-parity fertility response is concentrated; husband employment is stable or higher.

Consistency Check — Stated Ideal Fertility, by Subgroup

A labor market shock should move *both* realised parity and *stated ideal* n in the same subgroups and at the same thresholds.

Cross-sectional 2SLS of stated ideal n_i^* (and threshold cuts $\mathbb{1}\{n_i^* \geq k\}$) on $\bar{R}_{j(i),t(i)}$, ever-married four-wave sample. Same controls / FE / trends as the fertility specification.

Subgroup	Primary	Threshold decomposition			
	Ideal n	Ideal ≥ 2	Ideal ≥ 3	Ideal ≥ 4	Ideal ≥ 5
Full sample (pooled)	1.197** (0.541)	0.046 (0.061)	0.174 (0.254)	0.514* (0.270)	0.256** (0.109)
Lower education	1.533*** (0.549)	0.001 (0.088)	0.203 (0.247)	0.676*** (0.247)	0.351*** (0.101)
Arabic / Kurdish mother tongue	3.034** (1.253)	0.129 (0.158)	0.512 (0.417)	0.911*** (0.350)	0.475*** (0.184)
High-exposure border region	1.566** (0.734)	0.069 (0.117)	0.212 (0.237)	0.727** (0.336)	0.368** (0.143)

Stated ideal n shifts up at the upper thresholds (≥ 4 , ≥ 5), strongest in the same Arabic/Kurdish subgroup that carries the realised higher-parity result, with the same direction in lower-educ and the high-exposure border region. Internally consistent with the labor-market interpretation.

Higher-parity response concentrated among Arabic/Kurdish mother-tongue women; suggestive evidence in two related subgroups.

- **No effect** on marriage or first births anywhere.
- Pooled higher-parity (4–5) raw estimates are large and positive but do not survive Romano–Wolf across the 32-test family.
- Cells reject under Romano–Wolf at $\alpha = 0.05$ only in the **Arabic/Kurdish mother-tongue** subgroup, on three higher-parity cells. The **lower-educated** and **high-exposure East/Southeast** cuts show the same raw pattern but no cell rejects after Romano–Wolf — suggestive.
- **Hazard design** (complementary): overall null on every transition; the one effect robust to Romano–Wolf is a higher **second birth hazard for Kurdish/Arabic women** (RW $p = .03$) at ages 15 to 24: the second child arrives **earlier**, but completed family size **does not grow**.
- **Mechanism**: gender-asymmetric labor-market adjustment (Aksu, Erzan, Kırdar 2022) — female employment falls in exactly the subgroups where higher-parity fertility responds, husband employment stable or up. Lower opportunity cost without an income hit.
- **Consistency check**: stated ideal- n shifts up at the same upper thresholds in the same subgroups — supports the labor-market reading, not a separate channel.

Thank you. Questions?

Appendix

Appendix: Mean Age of Fertility Outcomes — Turkish vs. Syrian

Variable	Turkish			Syrian		
	Mean	SD	Obs	Mean	SD	Obs
Age at 1st Marriage	20.54	4.23	12,320	18.52	4.09	1,849
Age at 1st Birth	22.06	4.39	11,354	19.98	3.99	1,684
Age at 2nd Birth	24.88	4.64	9,068	21.80	3.92	1,393
Age at 3rd Birth	27.11	4.83	4,938	24.05	3.88	1,050
Age at 4th Birth	28.07	4.76	2,359	26.39	4.09	750
Age at 5th Birth	29.09	4.58	1,211	28.48	4.17	506

Source: TDHS 2013 and 2018; Syrian refugees use only the 2018 round (first nationally representative sample in the TDHS framework). Syrian women marry ~ 2 years earlier and reach each parity $\sim 2\text{--}3$ years younger than Turkish natives.

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Appendix: Fertility and Marital Status by Age Group

Age Group	Mean #Children	Median	Portion Married	Obs
<i>Panel A: Turkish women</i>				
12–16	0.001	0	0.010	967
17–21	0.175	0	0.214	2,441
22–26	0.904	1	0.631	2,375
27–30	1.671	2	0.878	1,974
31–35	2.306	2	0.940	2,681
36–40	2.827	2	0.960	2,521
41–45	2.952	3	0.964	2,157
46–49	3.100	3	0.963	1,594
Total	1.825	2	0.737	16,710
<i>Panel B: Syrian women</i>				
12–16	0.109	0	0.212	156
17–21	0.927	1	0.754	480
22–26	1.913	2	0.901	426
27–30	2.941	3	0.928	320
31–35	3.735	4	0.927	317
36–40	4.888	5	0.926	242
41–45	5.418	5	0.935	184
46–49	6.011	6	0.912	91
Total	2.766	2	0.834	2,216

Source: TDHS 2013 and 2018; Syrian refugees use only the 2018 round (first nationally representative sample in the TDHS framework). At every age window, Syrian women report higher mean parity and a higher married share than Turkish natives.

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The problem. The main fertility table is a family of **32 hypotheses** (5 parities $\times$ 6 ages + 2 marriage cells). At nominal $\alpha = 0.05$, ~ 1.6 false rejections are expected by chance even if no real effect exists.

What RW does. Adjusts each p -value so that the probability of *any* false rejection across the whole family stays below α – the **familywise error rate** (FWER):

$$\text{FWER} = \Pr(\text{at least one false rejection}) \leq \alpha.$$

How. Bootstrap the joint distribution of t -statistics – preserving province-level clustering and cross-cell correlation – and adjust each cell's p -value to the bootstrap probability that the most extreme remaining t -stat exceeds the observed one (**stepdown**).

What changes in the tables. Cells with raw $p < 0.05$ may have RW $p > 0.05$ – the bracketed values rise; this is the price of testing 32 hypotheses simultaneously.

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Appendix ● Four-wave EM sample

Main + three subgroups, TDHS 2003/08/13/18, childhood province.

Appendix: Main IV — Four-Wave Ever-Married Sample (RW)

Main IV (2SLS) results, ever-married sample.

Outcome	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
Ever Gave Birth to One Child	0.313 (1.914) [1.00]	-1.348 (1.229) [0.98]	0.986 (2.238) [1.00]	-0.608 (1.109) [1.00]	-1.397 (1.437) [0.99]	1.373** (0.680) [0.74]
Ever Gave Birth to Two Children	0.322* (0.172) [0.81]	1.273 (2.240) [1.00]	0.687 (1.452) [1.00]	-1.075 (2.018) [1.00]	2.253 (2.615) [0.99]	0.100 (1.927) [1.00]
Ever Gave Birth to Three Children	0.143 (0.117) [0.96]	1.482 (0.974) [0.91]	-0.772 (0.924) [0.99]	-0.900 (1.991) [1.00]	0.714 (2.365) [1.00]	-0.561 (2.507) [1.00]
Ever Gave Birth to Four Children	— (—)	— (—)	-0.368 (0.518) [1.00]	0.401 (0.754) [1.00]	0.029 (1.582) [1.00]	2.742 (2.004) [0.94]
Ever Gave Birth to Five Children	— (—)	— (—)	0.280 (0.305) [0.99]	0.789 (0.551) [0.92]	-0.346 (1.274) [1.00]	1.567 (1.624) [0.99]
Observations	19,281	21,525	21,782	19,776	16,442	12,282

Ever-married, TDHS 2003/08/13/18, childhood province. Trade-off: more pre-shock anchoring but (i) marriage is treatment-responsive (selection); (ii) measurement error on province.

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Appendix: Lower-Educated Women — Four-Wave EM Sample

Main IV (2SLS) results, ever-married sample.

Outcome	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
<i>Lower-educated women ($\leq$ primary)</i>						
Ever Birth ≥ 1	-0.452 (2.179) [1.00]	-1.843 (1.977) [0.99]	-0.507 (1.713) [1.00]	0.527 (1.357) [1.00]	0.251 (0.958) [1.00]	1.277 (0.888) [0.91]
Ever Birth ≥ 2	0.145 (0.237) [1.00]	0.151 (2.977) [1.00]	-1.999 (2.432) [1.00]	0.419 (1.279) [1.00]	1.410 (2.445) [1.00]	1.543 (1.342) [0.97]
Ever Birth ≥ 3	0.002 (0.154) [1.00]	1.480 (1.370) [0.97]	-1.751 (1.414) [0.97]	0.858 (2.529) [1.00]	1.391 (1.985) [1.00]	1.931 (2.720) [1.00]
Ever Birth ≥ 4	— (—)	— (—)	-0.605 (0.858) [1.00]	1.699 (1.219) [0.95]	1.063 (1.205) [0.99]	4.051 (2.468) [0.84]
Ever Birth ≥ 5	— (—)	— (—)	0.350 (0.412) [0.99]	1.148 (0.882) [0.96]	0.151 (1.468) [1.00]	2.733* (1.749) [0.84]
Observations	14,711	16,562	16,952	15,800	13,465	10,305

Positive higher-parity raw estimates at ages 28+ persist; no cell survives RW in the EM family.

Appendix: Kurdish / Arabic Mother Tongue — Four-Wave EM Sample

Main IV (2SLS) results, ever-married sample.

Outcome	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
<i>Arabic / Kurdish mother tongue</i>						
Ever Birth ≥ 1	1.473 (1.219) [0.92]	-1.001 (2.029) [1.00]	-1.113 (2.688) [1.00]	-2.204 (2.129) [0.95]	1.714 (1.892) [0.96]	0.322 (1.488) [1.00]
Ever Birth ≥ 2	0.251 (0.412) [1.00]	2.834* (1.771) [0.55]	2.140 (3.603) [1.00]	0.810 (2.459) [1.00]	2.998 (2.431) [0.90]	2.097* (1.582) [0.90]
Ever Birth ≥ 3	-0.082 (0.329) [1.00]	2.023 (1.993) [0.92]	-1.391 (1.874) [1.00]	12.349*** (2.473) [0.05]	3.253 (3.036) [0.96]	1.314 (2.216) [1.00]
Ever Birth ≥ 4	— (—)	— (—)	-0.184 (1.792) [1.00]	8.438** (2.470) [0.43]	2.087 (3.854) [1.00]	7.109*** (1.750) [0.17]
Ever Birth ≥ 5	— (—)	— (—)	0.306 (0.985) [1.00]	4.181 (2.658) [0.90]	6.160* (2.870) [0.75]	6.020** (2.091) [0.32]
Observations	4,492	4,765	4,517	3,885	3,041	2,157

The parity $\geq 3@28$ cell rejects at $\alpha = 0.05$ even in the EM sample ($p^{\text{RW}} \approx 0.05$). Higher-parity raw estimates at 4@28, 4@36, 5@28 are large but do not survive RW in the EM family.

Appendix: High-Exposure Border Region — Four-Wave EM Sample

Main IV (2SLS) results, ever-married sample.

Outcome	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
<i>High-exposure border region (E/SE)</i>						
Ever Birth ≥ 1	3.464** (1.693) [0.36]	-0.142 (1.035) [1.00]	-0.700 (0.997) [0.97]	1.105 (1.828) [0.96]	1.292 (1.170) [0.85]	0.781 (0.982) [0.92]
Ever Birth ≥ 2	0.723 (0.312) [0.72]	4.920*** (2.204) [0.17]	3.907*** (2.012) [0.39]	-0.081 (1.860) [1.00]	6.482*** (2.197) [0.17]	0.783 (1.061) [0.96]
Ever Birth ≥ 3	0.226 (0.182) [0.89]	1.177 (1.303) [0.88]	-0.264 (1.378) [1.00]	4.832* (2.055) [0.54]	9.565*** (2.461) [0.06]	4.237* (1.606) [0.60]
Ever Birth ≥ 4	— (—)	— (—)	0.568 (1.068) [0.97]	3.187 (1.430) [0.66]	7.355*** (1.890) [0.10]	8.343 (2.308) [0.60]
Ever Birth ≥ 5	— (—)	— (—)	0.331 (0.587) [0.97]	2.436** (1.102) [0.54]	4.365 (1.601) [0.64]	7.321** (3.818) [0.31]
Observations	5,666	6,103	5,823	4,995	3,954	2,796

Raw higher-parity point estimates at ages 28+ remain large and positive, but **no EM cell survives RW at $\alpha = 0.05$.**

Appendix ● Baseline parity-attainment by subgroup

Share of women in each age column who have reached the row outcome.

Appendix: Baseline Parity-Attainment — Lower-Educated Women

Outcome	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
Ever Married	16.1%	61.5%	85.5%	93.5%	96.0%	97.4%
Ever Gave Birth to One Child	5.4%	44.7%	76.1%	87.9%	92.6%	94.5%
Ever Gave Birth to Two Children	0.7%	15.4%	47.1%	69.9%	81.4%	86.5%
Ever Gave Birth to Three Children	0.0%	2.9%	16.0%	32.5%	46.7%	53.6%
Ever Gave Birth to Four Children	0.0%	0.7%	5.5%	14.2%	21.2%	26.6%
Ever Gave Birth to Five Children	0.0%	0.1%	1.6%	6.3%	10.8%	14.4%
Observations	8,316	8,514	8,468	7,739	6,477	5,026

Higher baseline higher-parity rates than the overall sample (e.g. parity ≥ 4 by 32: 21.2% vs. 15.9% overall) — room to move up the parity ladder.

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Outcome	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
Ever Married	16.6%	55.0%	78.5%	89.6%	93.6%	96.4%
Ever Gave Birth to One Child	5.5%	41.3%	69.7%	84.0%	90.5%	93.6%
Ever Gave Birth to Two Children	1.2%	20.4%	54.0%	74.1%	84.6%	90.9%
Ever Gave Birth to Three Children	0.1%	6.0%	28.8%	53.6%	69.4%	78.7%
Ever Gave Birth to Four Children	0.0%	1.8%	14.1%	33.9%	48.3%	60.6%
Ever Gave Birth to Five Children	0.0%	0.3%	4.7%	18.9%	31.8%	43.8%
Observations	2,940	2,722	2,396	1,987	1,526	1,090

Substantially higher baseline higher-parity rates (e.g. parity ≥ 4 by 32: 44.2% vs. 15.9% overall; parity ≥ 5 by 32: 28.7% vs. 8.0%). Largest scope for upward parity progression.

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Appendix: Baseline Parity-Attainment — High-Exposure Border Region

Outcome	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
Ever Married	12.9%	50.6%	75.3%	87.7%	92.9%	95.4%
Ever Gave Birth to One Child	4.6%	36.5%	65.5%	82.0%	89.6%	92.4%
Ever Gave Birth to Two Children	0.9%	17.3%	48.1%	69.6%	81.6%	88.2%
Ever Gave Birth to Three Children	0.1%	4.8%	24.5%	48.8%	64.8%	74.9%
Ever Gave Birth to Four Children	0.0%	1.4%	11.9%	29.7%	44.2%	58.2%
Ever Gave Birth to Five Children	0.0%	0.3%	3.8%	15.9%	28.7%	41.0%
Observations	3,757	3,406	2,981	2,453	1,902	1,350

Highest baseline higher-parity rates (e.g. parity ≥ 5 by 32: 28.7% – shared with the Kurdish/Arabic subgroup, since the two cuts overlap heavily in the East/Southeast). Common thread across all three heterogeneity slides: **effects appear where the parity ladder has room left to climb.**

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Appendix • Robustness

One slide per check; each returns to slide 10 (Main Results).

NUTS-1 trends

NUTS-2 trends

Drop post-2011 movers

4-wave decomposition

OLS counterpart

Event study (pre-trends)

Hazard: ever-married sample

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Appendix: Robustness — NUTS-1-Region Linear Trends

Outcome	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
Ever Married	-0.122 (0.247) [0.98]	-0.510 (0.683) [0.95]	0.414 (1.832) [0.98]	0.230 (1.070) [0.98]	0.859 (0.980) [0.93]	-0.413 (1.274) [0.98]
Ever Gave Birth to One Child	-0.020 (0.147) [1.00]	0.305 (0.587) [1.00]	-0.027 (2.139) [1.00]	1.051 (1.342) [1.00]	-1.000 (1.828) [1.00]	0.381 (1.528) [1.00]
Ever Gave Birth to Two Children	0.029 (0.050) [1.00]	0.359 (0.591) [1.00]	-0.126 (0.806) [1.00]	0.628 (2.184) [1.00]	1.726 (3.593) [1.00]	-0.883 (1.220) [1.00]
Ever Gave Birth to Three Children	0.004 (0.006) [1.00]	0.446** (0.224) [0.73]	-0.382 (0.595) [1.00]	0.769 (1.670) [1.00]	2.722 (2.679) [0.98]	1.327 (1.767) [1.00]
Ever Gave Birth to Four Children	— (—)	— (—)	0.557 (0.403) [0.93]	1.731** (0.849) [0.72]	1.647 (1.471) [0.97]	4.184* (2.159) [0.73]
Ever Gave Birth to Five Children	— (—)	— (—)	0.304 (0.241) [0.95]	1.601* (0.919) [0.82]	1.108 (0.911) [0.95]	4.020** (1.965) [0.72]
Observations	14,354	13,484	12,241	10,240	8,245	6,110

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Appendix: Robustness — NUTS-2-Region Linear Trends

Outcome	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
Ever Married	-0.288 (0.289) [0.84]	-1.093 (0.802) [0.68]	-0.758 (1.920) [0.90]	-1.000 (1.010) [0.84]	-0.129 (1.139) [0.90]	-0.757 (1.184) [0.90]
Ever Gave Birth to One Child	-0.107 (0.182) [1.00]	-0.264 (0.771) [1.00]	-1.124 (2.515) [1.00]	0.022 (1.436) [1.00]	-2.168 (1.822) [0.95]	-0.729 (1.575) [1.00]
Ever Gave Birth to Two Children	0.017 (0.039) [1.00]	-0.100 (0.457) [1.00]	-0.811 (1.015) [0.99]	-1.065 (2.557) [1.00]	-0.480 (3.478) [1.00]	-2.465 (1.582) [0.87]
Ever Gave Birth to Three Children	0.000 (0.006) [1.00]	0.298 (0.214) [0.90]	-0.755 (0.761) [0.98]	0.744 (1.759) [1.00]	1.413 (2.806) [1.00]	1.705 (1.959) [0.99]
Ever Gave Birth to Four Children	– (–)	– (–)	0.243 (0.411) [1.00]	1.630** (0.831) [0.74]	0.778 (2.428) [1.00]	3.357 (2.636) [0.93]
Ever Gave Birth to Five Children	– (–)	– (–)	0.232 (0.227) [0.98]	1.295 (0.871) [0.89]	1.345 (1.194) [0.96]	3.523 (3.186) [0.96]
Observations	14,354	13,484	12,241	10,240	8,245	6,110

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Appendix: Robustness — Dropping Post-2011 Out-Migrants

	Parity ≥ 4		Parity ≥ 5	
	by 28	by 32	by 28	by 32
Primary sample	1.905** (0.809)	2.246 (1.415)	1.392* (0.803)	1.414 (0.912)
<i>N</i>	10,240	8,245	10,240	8,245
Drop post-2011 movers	3.220*** (1.101)	3.914* (2.059)	1.668 (1.177)	1.399 (1.026)
<i>N</i>	9,365	7,643	9,365	7,643

Concern: women in high- B_n provinces in 2011 may have moved away, attenuating exposure for those who stayed. Movers = 12.3% pooled (7.25% in 2013, 19.53% in 2018). Mover indicator **uncorrelated with** B_n ($\hat{\beta} = 0.002$, $p = 0.73$). Non-mover estimates are **larger and more precise** at the four core higher-parity cells, consistent with cleaner exposure measurement among stayers.

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Appendix: Robustness — Four-Wave Decomposition Ladder

Core outcome	(i) Primary 2013/18, full, prov. 2011	(ii) EM, prov. 2011	(iii) EM, childhood prov.	(iv) Four-wave 2003–18, EM, childhood prov.
Ever Gave Birth to Four Children, age 28	1.905** (0.809)	1.768** (0.879)	1.750* (0.940)	1.211 (1.007)
Ever Gave Birth to Four Children, age 32	2.246 (1.415)	2.260 (1.375)	2.098 (1.543)	1.198 (1.517)
Ever Gave Birth to Five Children, age 28	1.392* (0.803)	1.313 (0.851)	0.747 (0.749)	1.228 (0.997)
Ever Gave Birth to Five Children, age 32	1.414 (0.912)	1.437 (0.924)	1.159 (1.238)	1.099 (1.067)
Observations (egb4, age 28)	10,240	9,746	9,746	19,776

Decomposes the gap between primary FS estimates and four-wave EM estimates by changing one feature at a time.

- (i)→(ii): ever-married restriction (small effect).
- (ii)→(iii): replacing 2011 province with childhood province (**attenuates parity 5@28 from 1.31 to 0.75** — measurement error).
- (iii)→(iv): adds 2003/08 waves (further attenuates; less precision).

Reading: **FS spec is preferred baseline**; four-wave EM is a complementary lower-bound check.

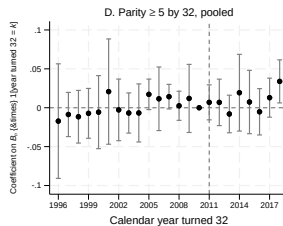
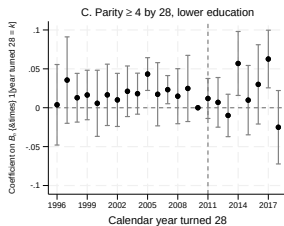
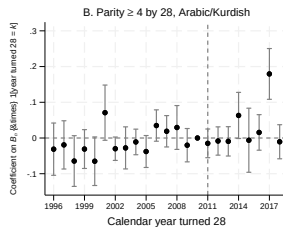
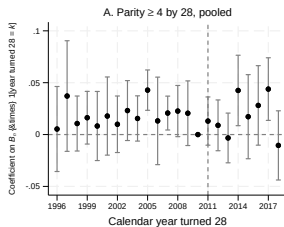
Appendix: Robustness — OLS Counterpart of the Main 2SLS

Outcome	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
Ever Married	-0.036 (0.176)	-0.743 (0.635)	0.799 (1.472)	-1.056 (1.458)	-0.414 (0.758)	-0.164 (0.862)
Ever Gave Birth to One Child	-0.013 (0.131)	-0.259 (0.418)	0.655 (1.327)	-0.185 (1.561)	-1.190 (1.145)	-0.478 (0.975)
Ever Gave Birth to Two Children	0.036 (0.029)	0.443 (0.391)	0.606 (0.788)	0.320 (1.584)	0.080 (2.349)	-0.220 (1.091)
Ever Gave Birth to Three Children	0.006* (0.003)	0.128 (0.319)	-0.516 (0.429)	1.592 (1.387)	2.069 (2.045)	2.311* (1.369)
Ever Gave Birth to Four Children	— (—)	— (—)	0.050 (0.263)	1.367 (0.909)	2.565** (1.130)	3.543** (1.543)
Ever Gave Birth to Five Children	— (—)	— (—)	0.020 (0.353)	0.808 (0.718)	2.888*** (0.742)	1.359 (1.301)
Observations	14,354	13,484	12,241	10,240	8,245	6,110

OLS estimates on the same sample / controls / trends as the main 2SLS. **Coefficients are close in magnitude to the 2SLS:** the four core higher-parity cells differ by less than ~ 1.5 between OLS and 2SLS, and the sign / pattern is unchanged. The IV machinery is correcting selection-on-province bias but is **not generating the headline pattern**; the same higher-parity signature appears in OLS.

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Appendix: Robustness — Event Study around 2011



Appendix: Hazard — Ever Married Four Wave Sample

Same hazard specification (NUTS-2 + MT trends, contemporaneous ratio instrumented) on the **ever-married four-wave sample** (TDHS 2003/08/13/18, childhood province; birth transitions only).

	1st birth	2nd birth	3rd birth	4th birth	5th birth
2SLS $\hat{\beta}$	-0.28*** (0.10)	0.21 (0.28)	0.41*** (0.13)	0.13 (0.11)	0.10 (0.12)
[RW p]	[0.18]	[0.68]	[0.15]	[0.64]	[0.68]
person years	196,172	73,685	83,513	42,997	20,576

The cells significant on raw inference (first birth negative, third birth positive) **do not survive Romano–Wolf** (min RW $p = .15$), consistent with the overall null in the full sample. EM caveats as before: marriage selection and measurement error in childhood province.

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Appendix ● Setting, sample, data

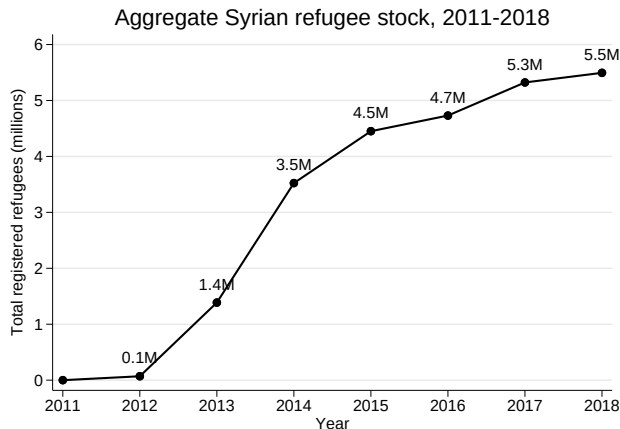
Appendix: First-Stage F and Coefficient on $\bar{Z}_{ja}$

	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
<i>Panel A: Primary sample (2013/18, full sample, province in 2011)</i>						
$\bar{Z}_{ja} (\times 10^{-5})$	2.514*** (0.379)	2.229*** (0.195)	2.269*** (0.224)	2.453*** (0.371)	2.293*** (0.315)	2.310*** (0.193)
First-stage F	43.9	131.3	102.4	43.7	52.9	143.7
Observations	14,354	13,484	12,241	10,240	8,245	6,110
<i>Panel B: Four-wave sample (2003/08/13/18, ever-married, childhood province)</i>						
$\bar{Z}_{ja} (\times 10^{-5})$	2.222*** (0.410)	2.359*** (0.411)	2.389*** (0.334)	2.430*** (0.368)	2.251*** (0.309)	2.338*** (0.256)
First-stage F	29.4	32.9	51.2	43.5	53.0	83.2
Observations	19,281	21,525	21,782	19,776	16,442	12,282

First-stage $F \in [43, 144]$ across all primary-sample cells; $F \in [29, 83]$ in the four-wave EM sample. Instrument is **strong** across the board; weak-IV concerns do not bind.

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Appendix: Aggregate Refugee Stock T_t across Host Countries



T_t builds up progressively after 2011 across Turkey, Lebanon, Jordan, and Iraq, rather than as a single step change. The shock activates B_n gradually over the 2011–2018 window.

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Primary sample (full): TDHS 2013 + 2018.

- Ever- and never-married women aged 15–49 at interview.
- Province assigned by **2011 residence** (pre-shock, predetermined).
- Full controls available (wave, childhood urban/rural, mother's education, mother tongue).

Four-wave EM (appendix): TDHS 2003 + 2008 + 2013 + 2018.

- Ever-married only (1998/2003/2008 surveyed only ever-married women).
- Province by **childhood province** (no detailed adult migration history in 2003/08).
- Used as complementary specification; subject to selection + measurement error.

Why FS leads. Marriage is itself a treatment-responsive outcome (selection); classical measurement error on province $\Rightarrow$ attenuation bias toward zero.

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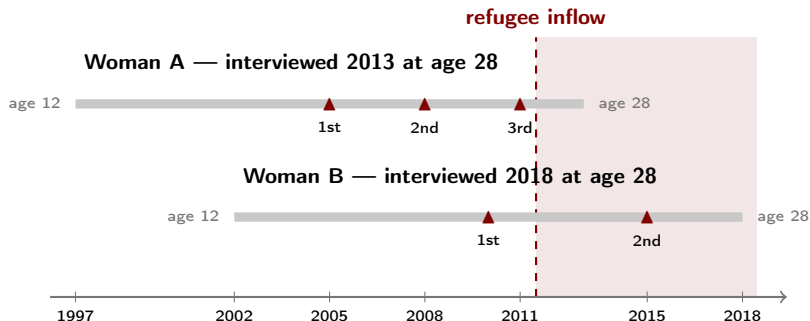
Province-year ratio. $R_{j,t}$ = registered Syrians / native population in province j , year t , from **Ministry of Interior** / **AFAD** administrative records (2012–2018; zero before 2012).

Cohort variation in $\bar{R}_{ja}$. A 2013 woman observed at age 28 averages exposure over 1997–2013 (**2 post-shock years**); a 2018 woman at age 28 averages over 2002–2018 (**7 post-shock years**). Same age, same province, *different* $\bar{R}_{ja}$. Crossed with B_j , this cross-cohort variation in window timing is what the IV uses on top of FE.

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Appendix: Worked Example — “Third Child by Age 28”

$Y_{i,28}^{(3)} = \mathbb{1}\{\text{parity} \geq 3 \text{ by age } 28\}$; the exposure $\bar{R}_{j,28}$ averages the province ratio over the calendar years in which the woman was 12–28.



	3 births by age 28?	window years post-2011	exposure
Woman A	$Y = 1$	2 of 17 (2012–13)	$\bar{R}_{j,28}$ low
Woman B	$Y = 0$	7 of 17 (2012–18)	$\bar{R}_{j,28}$ high

Appendix ● Complement subgroups

Panel B of paper heterogeneity tables: the other half of each cut.

Appendix: Higher-Educated Women

Outcome	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
<i>Higher education (secondary / tertiary)</i>						
Ever Married	0.018 (0.053) [0.99]	0.214 (0.589) [0.99]	-0.886 (3.014) [0.99]	-1.412 (3.269) [0.99]	-1.633 (2.920) [0.99]	1.927 (4.802) [0.99]
Ever Birth ≥ 1	0.041 (0.029) [0.94]	-0.391 (0.672) [0.99]	-0.579 (2.911) [1.00]	0.481 (3.708) [1.00]	-9.425* (4.929) [0.85]	0.625 (5.498) [1.00]
Ever Birth ≥ 2	0.001 (0.024) [1.00]	0.304 (0.239) [0.96]	0.274 (1.155) [1.00]	0.100 (4.655) [1.00]	-4.555 (7.553) [0.99]	-15.097* (8.562) [0.88]
Ever Birth ≥ 3	- (-) [0.96]	0.080 (0.061) [0.96]	0.044 (0.806) [1.00]	-3.229** (1.266) [0.67]	7.384 (6.983) [0.97]	-18.786** (7.432) [0.67]
Ever Birth ≥ 4	- (-) [0.97]	- (-) [0.97]	-0.571 (0.497) [0.97]	-0.322 (0.370) [0.98]	3.927 (3.189) [0.96]	-8.802* (4.863) [0.87]
Ever Birth ≥ 5	- (-) [0.97]	- (-) [0.97]	-0.486 (0.481) [0.97]	0.130 (0.186) [0.99]	-0.365 (0.567) [0.99]	-2.314 (2.160) [0.97]
Observations	4,993	4,352	3,512	2,607	1,897	1,213

Secondary / tertiary education. Raw point estimates are noisy and not concentrated at higher parities; no cell rejects under raw inference or under RW. Consistent with the lower-education subgroup carrying the higher-parity response.

Appendix: Turkish Mother Tongue

Outcome	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
<i>Turkish mother tongue</i>						
Ever Married	-0.433** (0.197) [0.37]	-1.286 (0.796) [0.53]	-0.143 (1.681) [0.98]	0.240 (1.621) [0.98]	-1.519 (1.295) [0.70]	-2.385 (2.949) [0.83]
Ever Birth ≥ 1	-0.172 (0.120) [0.94]	-0.813 (0.773) [0.98]	-0.569 (2.379) [1.00]	1.233 (1.928) [1.00]	-3.926 (2.778) [0.94]	-2.142 (3.325) [1.00]
Ever Birth ≥ 2	-0.003 (0.024) [1.00]	0.062 (0.747) [1.00]	-1.223 (0.906) [0.94]	0.191 (2.764) [1.00]	-0.779 (5.461) [1.00]	-3.663 (2.398) [0.92]
Ever Birth ≥ 3	0.004 (0.004) [0.99]	0.165 (0.317) [1.00]	-0.343 (0.570) [1.00]	-1.099 (1.785) [1.00]	2.768 (4.006) [1.00]	-0.849 (3.397) [1.00]
Ever Birth ≥ 4	– (–)	– (–)	-0.075 (0.325) [1.00]	-0.346 (1.064) [1.00]	2.582** (1.061) [0.70]	1.821 (2.959) [1.00]
Ever Birth ≥ 5	– (–)	– (–)	0.261** (0.115) [0.74]	-0.043 (0.310) [1.00]	0.026 (1.540) [1.00]	5.116 (3.823) [0.94]
Observations	11,013	10,512	9,655	8,137	6,634	4,960

Turkish mother-tongue women. Point estimates near zero across parities and ages; **no raw or RW rejection at higher parities**. The Kurdish/Arabic subgroup is the one that drives the higher-parity result.

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Appendix: Rest of Turkey

Outcome	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
<i>Rest of Turkey</i>						
Ever Married	-0.261 (0.187) [0.54]	-1.358 (1.073) [0.56]	1.726 (1.784) [0.56]	0.047 (1.257) [0.97]	-2.056 (1.606) [0.56]	-1.905** (0.754) [0.10]
Ever Birth ≥ 1	0.045 (0.226) [1.00]	-0.394 (1.020) [1.00]	1.823 (1.400) [0.88]	1.304 (2.049) [0.99]	-6.070** (2.517) [0.56]	-0.989 (0.978) [0.96]
Ever Birth ≥ 2	-0.036 (0.043) [0.98]	-0.595 (0.616) [0.97]	-1.202 (0.967) [0.89]	2.665 (2.110) [0.89]	-4.895 (3.171) [0.80]	-1.143 (1.641) [0.99]
Ever Birth ≥ 3	-0.001 (0.001) [1.00]	0.185 (0.348) [1.00]	-0.654 (0.790) [0.98]	-0.135 (3.917) [1.00]	0.192 (3.875) [1.00]	0.594 (2.686) [1.00]
Ever Birth ≥ 4	— (—)	— (—)	0.227 (0.478) [1.00]	1.448 (1.377) [0.95]	1.494 (2.367) [0.99]	6.100** (2.545) [0.56]
Ever Birth ≥ 5	— (—)	— (—)	0.580** (0.270) [0.61]	0.892 (1.347) [0.99]	1.485 (1.636) [0.98]	5.437* (2.798) [0.65]
Observations	10,352	9,962	9,219	7,836	6,394	4,799

Provinces outside the East/Southeast high-exposure region. Point estimates small and mostly insignificant; the headline higher-parity response is **absent** away from the heavily-treated border region.

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Appendix: Placebo with NUTS-1 Trends

Outcome	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
Ever Married	0.0004 (0.0005) [0.77]	0.0013 (0.0008) [0.54]	0.0016* (0.0010) [0.54]	0.0012** (0.0005) [0.38]	0.0014* (0.0008) [0.54]	-0.0004 (0.0007) [0.77]
Ever Gave Birth to One Child	0.0006** (0.0003) [0.72]	0.0014 (0.0010) [0.89]	0.0012 (0.0013) [0.98]	0.0013** (0.0006) [0.69]	0.0017** (0.0008) [0.72]	0.0002 (0.0007) [1.00]
Ever Gave Birth to Two Children	0.0001 (0.0002) [1.00]	-0.0000 (0.0005) [1.00]	0.0015* (0.0008) [0.76]	0.0025*** (0.0009) [0.54]	0.0024** (0.0011) [0.72]	0.0006 (0.0012) [1.00]
Ever Gave Birth to Three Children	0.0000 (0.0000) [0.94]	0.0001 (0.0003) [1.00]	0.0005 (0.0006) [0.99]	0.0002 (0.0007) [1.00]	-0.0000 (0.0007) [1.00]	-0.0024 (0.0014) [0.81]
Ever Gave Birth to Four Children	- (-)	- (-)	0.0005 (0.0004) [0.91]	0.0011 (0.0010) [0.94]	0.0009 (0.0024) [1.00]	-0.0012 (0.0028) [1.00]
Ever Gave Birth to Five Children	- (-)	- (-)	0.0000 (0.0002) [1.00]	0.0004 (0.0005) [0.99]	0.0013 (0.0015) [0.99]	0.0009 (0.0029) [1.00]
Observations	12,355	11,560	10,226	8,194	6,063	4,098

Same placebo specification as the main placebo slide ($Y \sim B_n \times t$ on $t(a, i) < 2012$) but with **NUTS-1-region** linear trends in place of macro-region trends. Coefficients remain near zero; **no RW rejection**.

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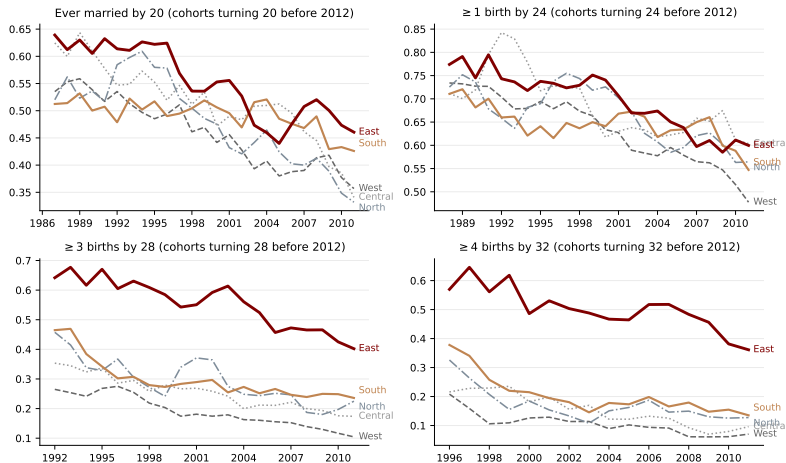
Appendix: Placebo with NUTS-2 Trends

Outcome	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
Ever Married	-0.0004 (0.0005) [0.75]	0.0017 (0.0011) [0.45]	-0.0016 (0.0010) [0.44]	-0.0006 (0.0008) [0.75]	-0.0011 (0.0013) [0.75]	-0.0034** (0.0016) [0.26]
Ever Gave Birth to One Child	0.0010** (0.0004) [0.58]	0.0015 (0.0013) [0.95]	-0.0009 (0.0015) [0.99]	-0.0009 (0.0009) [0.95]	-0.0007 (0.0014) [0.99]	-0.0021 (0.0022) [0.95]
Ever Gave Birth to Two Children	0.0003 (0.0003) [0.95]	-0.0010 (0.0010) [0.95]	0.0010 (0.0015) [0.99]	0.0002 (0.0008) [0.99]	-0.0021* (0.0011) [0.73]	-0.0012 (0.0014) [0.97]
Ever Gave Birth to Three Children	0.0000 (0.0000) [0.99]	-0.0002 (0.0004) [0.99]	-0.0004 (0.0009) [0.99]	-0.0007 (0.0015) [0.99]	-0.0017 (0.0016) [0.95]	-0.0067* (0.0035) [0.69]
Ever Gave Birth to Four Children	– (–)	– (–)	0.0004 (0.0006) [0.99]	0.0022 (0.0019) [0.95]	0.0028 (0.0041) [0.99]	-0.0022 (0.0049) [0.99]
Ever Gave Birth to Five Children	– (–)	– (–)	-0.0001 (0.0003) [0.99]	0.0010 (0.0010) [0.95]	0.0037 (0.0024) [0.83]	0.0050 (0.0043) [0.94]
Observations	12,355	11,560	10,226	8,194	6,063	4,098

Same placebo specification as the main placebo slide but with **NUTS-2-region** (subregion) linear trends. Coefficients remain near zero across all cells; **no RW rejection** – pre-trend null is robust to finer trend structures.

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Appendix: Regional Pre-Trends in Our Outcomes



Same construction for four cells of our outcome grid (cohorts before 2012, weighted, smoothed). The **East** (maroon) is above and on its own trajectory in every panel.

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Appendix: Hazard Placebo — Pre-2012 Person Years

Before 2012 the contemporaneous ratio is ≈ 0 , so any pre-period “effect” can only load on the settlement propensity: regress each transition’s event on $B_n \times t$ using **pre-2012 at risk person years** only. A valid design requires nulls everywhere.

	Marriage	1st birth	2nd birth	3rd birth	4th birth	5th birth
$\hat{\gamma} (B_n \times t)$	0.0001 (0.0002)	0.0003 (0.0003)	0.0002 (0.0006)	−0.0008 (0.0006)	0.0014 (0.0010)	0.0016 (0.0013)
[RW p]	[0.59]	[0.60]	[0.79]	[0.60]	[0.60]	[0.60]
person years	105,366	124,589	32,428	33,535	15,848	6,722

Nothing moves — no transition is significant, raw or after Romano–Wolf. The NUTS-2+MT trend set is the one chosen *by* this placebo: coarser trend sets leave a differential second birth trend before 2012. Within the Kurdish/Arabic subgroup the placebo also yields **no rejection at 5%** after RW (min RW $p = .06$).

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Appendix: Kurdish/Arabic Second Birth Hazard by Age Band

Locating the second birth response: same hazard IV, women at risk (exactly one child), split by five year age band.

	15–19	20–24	25–29	30–34
2SLS $\hat{\beta}$	2.29*** (0.72)	1.72*** (0.49)	0.62 (2.26)	–0.85 (0.87)
person years	1,078	2,596	1,174	452

Band level raw inference; bands 35+ omitted (<200 at risk person years). The response is **concentrated among young women** (15 to 24) and absent after 25: the second child comes **earlier in life**, with **no rise in births overall**.

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Appendix • Comparison with Aboulhosn, Aksoy & Özcan

Appendix: Comparison with Aboulhosn, Aksoy & Özcan (AAO)

Aboulhosn, Aksoy & Özcan: *Refugee Immigration and Natives' Fertility*.

- **Same research question:** how do Syrian refugees affect Turkish native fertility?
- **Different data:** they use the National Survey on Domestic Violence Against Women (**NSDVW**) 2008 and 2014; we use TDHS waves with a pseudo-panel structure.
- **Different design:** cross-sectional DiD on two waves vs. our continuous-DiD on an extended pseudo-panel.
- **Different outcome structure:** we track parity-by-age progression $\mathbb{1}\{\text{parity} \geq p \text{ by age } a\}$; they use contemporaneous 1-year birth/pregnancy indicators.

AAO headline IV results (Table 1).

Outcome	AAO IV
Gave birth past year	0.318**
Currently pregnant	0.248**
Number of children	1.870**
Birth & not 1st child	0.305***
Birth & 1st child	0.012
Pregnant & ≥ 1 child	0.317**

Existing mothers, not childless women, drive the response.

Appendix: AAO Table 1 — A Three-Step Narrative

AAO's data (NSDVW 2008+2014); IV coefficients on refugee/pop. ratio.

	AAO	1-to-1 repl.	sample fixed	+ region trends
Gave birth past year	0.318** (0.166)	0.384 (0.247)	0.278 (0.244)	-0.117 (0.322)
Currently pregnant	0.248** (0.112)	0.388** (0.171)	0.435** (0.170)	0.447** (0.178)
Number of children	1.870** (0.928)	2.004* (1.067)	2.084* (1.175)	2.818*** (1.085)
Birth & not 1st	0.305*** (0.083)	0.315* (0.164)	0.268** (0.132)	-0.033 (0.161)
Birth & 1st child	0.012 (0.137)	0.069 (0.176)	0.011 (0.153)	-0.085 (0.183)
Pregnant & ≥ 1 child	0.317** (0.164)	0.381 (0.247)	0.376** (0.146)	0.420*** (0.133)
Pregnant no children	0.000 (0.006)	0.003 (0.003)	0.059 (0.096)	0.027 (0.100)
Sample <i>N</i>	11,285	11,285	14,058	14,058

- **1-to-1 repl.** On their data, their spec — we reproduce their headline within rounding.
- **Sample fixed.** AAO's setup drops women on three filters (asymmetric 2008 schooling recode; missing birth-month/year; inconsistent missing-pregnancy handling) shrinking *N* from 14,058 to 11,285.
- **+ region trends.** Adding region $\times$ year trends drives "Gave birth past year" and "Birth & not 1st child" to zero.

Supplementary check, not the main critique. The main critique is the 3-step narrative on the previous slide. Here we re-run AAO's exact specification (same controls, same IV, same outcomes) on a different Turkish dataset — our **TDHS 2008 + 2013** ever-married 15–49 sample ($N = 14,621$).

Outcome	AAO paper IV	TDHS replication IV
Gave birth past year	0.318**	0.301
Currently pregnant	0.248**	−0.465**
Number of children	1.870**	2.588*
Birth & not 1st child	0.305***	−0.121
Birth & 1st child	0.012	0.421*
Pregnant & ≥ 1 child	0.317**	−0.223

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